Mixtilinear Incircles

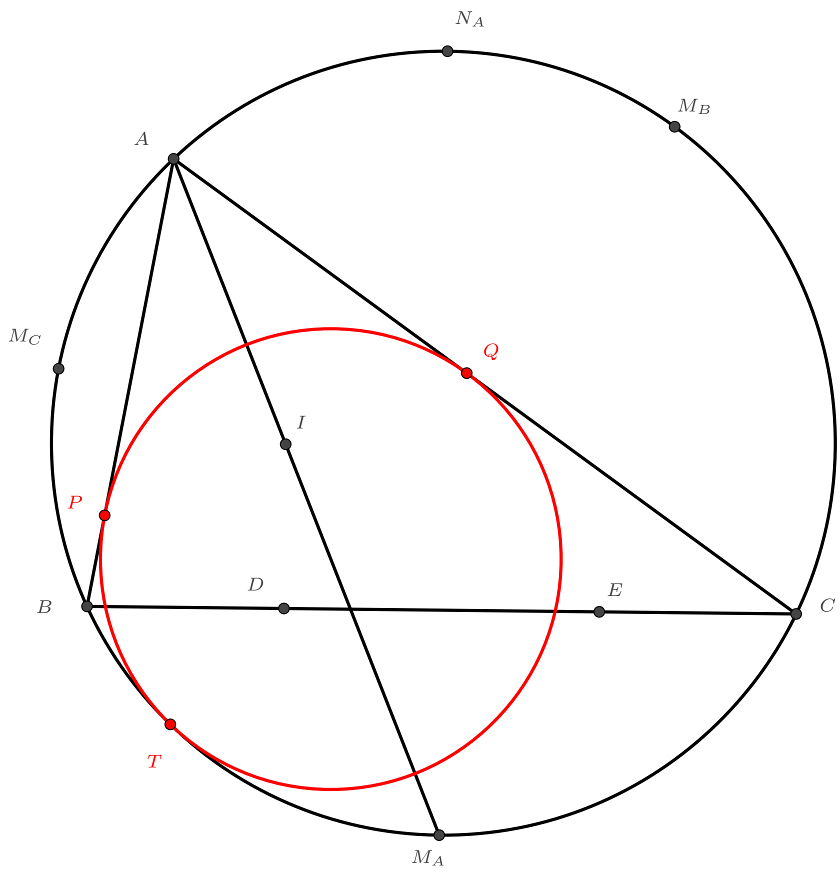
Mandy Kwok

January 30, 2021

Throughout the notes, (XYZ) denotes the circumcircle of $\triangle XYZ$.

In $\triangle ABC$, the **MIXTILINEAR INCIRCLE** is the circle in red, touching AB, AC and the circumcircle of $\triangle ABC$ at P, Q and T respectively.

In the figure below, I is the incentre, M_A, M_B, M_C are the midpoints of arc BC, CA, AB not containing A, B, C. N_A is the midpoint of arc BAC, D and E are the points on BC touching the incircle and A-excircle respectively. Denote the A-mixtilinear incircle by ω.



We have the following properties:

1. T, P, M_C are collinear. (Similarly, T, Q, M_B are collinear.)
2. I is the midpoint of PQ.
3. T, I, N_A are collinear.
4. TI and AE meet at ω.
5. $\angle BAT = \angle CAE$.
6. $\angle ATB = \angle CTD$.
7. $\angle ATM_C = \angle M_BTI$.
8. B, P, I, T are concyclic. (Similarly, C, Q, I, T are concyclic.)
9. CI and M_CB are tangent to the circle $(BPIT)$.
10. T is the centre of spiral similarity sending BM_CI to IM_BC.
11. AI and BC meet at the circle (DTM_A).
12. BC, TM_A and PQ are concurrent.
13. TM_A and AD meet on ω.

Problems:

14. Define the A-mixtilinear excircle and derive the analogous properties alike the above.
15. In triangle ABC with incentre I, suppose the A-mixtilinear incircle touches the circumcircle at T. AT meets the incircle of $\triangle ABC$ at X and Y, where X is in between A and Y. Show that $AT \cdot AX = AI^2$.
16. In triangle ABC, let I be the incentre of $\triangle ABC$, and let AI meets (ABC) at M. Suppose the incircle of $\triangle ABC$ meets BC at D, and let Z be the midpoint of ID. Show that MZ is the radical axis of the B-mixtilinear incircle and the C-mixtilinear incircle of $\triangle ABC$.

Challenges:

17. (IMO 1999 G8) In triangle ABC, let Ω be its circumcircle. Let X be a variable point on the arc AB not containing C, and let O_1 and O_2 be the incentres of $\triangle CAX$ and $\triangle CBX$. Prove that (XO_1O_2) intersects Ω at a fixed point.
18. (USAMO TST 2016 P2) Let ABC be a scalene triangle, and suppose the incircle of ABC touches BC at D. The angle bisector of $\angle BAC$ meets BC and (ABC) at E and F. The circle (DEF) intersects Ω at $T \neq F$, and AT meets (DEF) again at $S \neq T$. Prove that S lies on the A-excircle of $\triangle ABC$.
19. (IMO 2014 G7) Let ABC be a triangle with incentre I. Let the line passing through I and perpendicular to CI intersect the segment BC and the arc BC (not containing A) of (ABC) at points U and V, respectively. Let the line passing through U and parallel to AI intersect AV at X, and let the line passing through V and parallel to AI intersect AB at Y . Let W and Z be the midpoints of AX and BC, respectively. Prove that if the points I, X, and Y are collinear, then the points I, W, and Z are also collinear.

References:

1. Evan Chen (2015). *A Guessing Game: Mixtilinear Incircles*. Personal blog of Evan Chen.
2. Jafet98 (2018). *On Mixtilinear Incircles*. Art of Problem Solving.